Lecture 43–44 – 50 Questions and Answers

Topic: Unsupervised Machine Learning with Python

I. Data Preparation and Overview (1–10)

1. **Q:** What dataset is used in the tutorial for unsupervised learning?
   A: The Iris dataset, containing 150 samples from three species of Iris flowers.

2. **Q:** How many features does the Iris dataset contain?
   A: Four features — sepal length, sepal width, petal length, and petal width.

3. **Q:** How many classes or species does the Iris dataset include?
   A: Three — Setosa, Versicolor, and Virginica.

4. **Q:** Why is standardization important before applying PCA or clustering?
   A: Because these methods are sensitive to feature scales; standardization ensures equal contribution from all features.

5. **Q:** What Python library is used for standardizing data?
   A: `sklearn.preprocessing.StandardScaler`.

6. **Q:** What is the typical split ratio for training and testing the dataset?
   A: 70% training and 30% testing.

7. **Q:** What Python library is used for splitting the dataset?
   A: `sklearn.model_selection.train_test_split`.

8. **Q:** Why is the scaler fitted only on training data?
   A: To prevent data leakage from test data into the training process.

9. **Q:** What shape did the training and testing sets have for the Iris dataset?
   A: Training: (105, 4), Testing: (45, 4).

10. **Q:** What visualization libraries are used in the tutorial?
    A: `matplotlib` and `seaborn`.

II. Principal Component Analysis (11–20)

11. **Q:** What is PCA primarily used for?
    A: Linear dimensionality reduction and visualization of high-dimensional data.

12. **Q:** What does PCA find in the dataset?
    A: The directions (principal components) of maximum variance.

13. **Q:** What are the main properties of principal components?
    A: They are orthogonal and ordered by explained variance.

14. **Q:** What does the explained variance ratio represent?
    A: The proportion of total dataset variance captured by each principal component.

15. **Q:** What was the explained variance ratio for the four components in the Iris dataset?
    A: [0.7399, 0.2151, 0.0396, 0.0052].

16. **Q:** What cumulative variance was captured by all four components?
    A: 100% (1.0 cumulative variance ratio).

17. **Q:** What is a scree plot used for in PCA?
    A: To visualize how much variance each component explains and to determine the number of components to retain.

18. **Q:** What heuristic is commonly used for selecting the number of principal components?
    A: Keep components until cumulative variance reaches 80–95% or the “elbow” in the scree plot.

19. **Q:** What insight is gained from PCA visualization plots?
    A: How well components separate different classes or capture data structure.

20. **Q:** What do later principal components typically represent?
    A: Noise or minor variations in the dataset.

III. k-means Clustering (21–30)

21. **Q:** What is the goal of k-means clustering?
    A: To partition data into k clusters minimizing within-cluster sum of squares (WCSS).

22. **Q:** What type of clusters does k-means usually form?
    A: Roughly spherical clusters of similar size.

23. **Q:** What method is used to find the optimal number of clusters?
    A: The elbow method.

24. **Q:** What is plotted in the elbow method?
    A: WCSS vs. the number of clusters.

25. **Q:** What is the optimal number of clusters found for the Iris dataset?
    A: k = 3.

26. **Q:** What does WCSS stand for?
    A: Within-Cluster Sum of Squares.

27. **Q:** What Python class implements k-means clustering?
    A: `sklearn.cluster.KMeans`.

28. **Q:** How are cluster centers (centroids) computed?
    A: By averaging the feature values of points assigned to each cluster.

29. **Q:** How are training and testing clusters distributed in the Iris example?
    A: Training clusters: [32, 33, 40]; Test clusters: [13, 17, 15].

30. **Q:** How can k-means results be visualized?
    A: In the original feature space and PCA-reduced space using scatter plots.

IV. DBSCAN Clustering (31–40)

31. **Q:** What does DBSCAN stand for?
    A: Density-Based Spatial Clustering of Applications with Noise.

32. **Q:** What makes DBSCAN different from k-means?
    A: It does not require specifying the number of clusters beforehand and can detect noise points.

33. **Q:** What are the two key parameters in DBSCAN?
    A: `eps` (maximum distance between neighbors) and `min_samples` (minimum points to form a cluster).

34. **Q:** What type of clusters can DBSCAN find?
    A: Clusters of arbitrary shape.

35. **Q:** What is the purpose of a k-distance graph in DBSCAN?
    A: To help estimate the optimal `eps` value by finding an "elbow."

36. **Q:** What value of `eps` was selected in the Iris dataset example?
    A: `eps = 1.25`.

37. **Q:** How many clusters and noise points did DBSCAN detect in the Iris dataset?
    A: Two clusters and one noise point.

38. **Q:** How is DBSCAN visualized in PCA space?
    A: Each cluster is represented by color, and noise points are shown separately.

39. **Q:** How does DBSCAN handle noise points?
    A: It assigns them the label `-1`.

40. **Q:** What happens if `eps` is set too small or too large?
    A: Too small → many small clusters; Too large → one large cluster.

V. Nonlinear Dimensionality Reduction (t-SNE and UMAP) (41–50)

41. **Q:** What does t-SNE stand for?
    A: t-Distributed Stochastic Neighbor Embedding.

42. **Q:** What is the primary purpose of t-SNE?
    A: To visualize high-dimensional data while preserving local structure.

43. **Q:** What is the key hyperparameter in t-SNE?
    A: Perplexity.

44. **Q:** What does perplexity roughly represent?
    A: The effective number of neighbors each point considers.

45. **Q:** What is a good range of perplexity values to test?
    A: Between 5 and 50.

46. **Q:** What library is used to implement t-SNE in Python?
    A: `sklearn.manifold.TSNE`.

47. **Q:** What does UMAP stand for?
    A: Uniform Manifold Approximation and Projection.

48. **Q:** How does UMAP compare to t-SNE?
    A: UMAP is faster, scales better, and preserves more global structure.

49. **Q:** What is the key parameter controlling neighborhood size in UMAP?
    A: `n_neighbors`.

50. **Q:** What values of `n_neighbors` were tested in the example?
    A: 5, 15, 30, and 50.